

①

$$a. \quad \frac{dA}{dv} = -\eta K'' S_A \rho_c (1-E) C_A$$

$$\int_{C_{A0}}^{C_A} \frac{dC_A v}{dv} = - \int_0^v \eta K'' S_A \rho_c (1-E) C_A$$

$$A = \frac{\eta K'' S_A \rho_c (1-E)}{v}$$

$$\ln \left[\frac{C_A}{C_{A0}} \right] = - \frac{\eta K'' S_A \rho_c (1-E)}{v} dv$$

$$\ln \left[\frac{C_A}{C_{A0}} \right] = -Av = -A \left(\frac{w}{\rho_c (1-E)} \right)$$

$$A = \frac{-\ln \left[\frac{C_A}{C_{A0}} \right] (1-E) \rho_c}{w} = \frac{-\ln (1-x) (1-E) \rho_c}{w}$$

$$A = \frac{-\ln (0.5) (1-0.4) (2.5 \text{ g/m}^3)}{500 \text{ g}} = 2.08 \times 10^{-3} \text{ 1/cm}^3$$

②

$$\begin{aligned} K_2'' &= a e^{-EA/RT} \\ &= (1 \times 10^5 \text{ cm/min}) \exp \left[\frac{-100000 \text{ J/mol}}{(8.314) (400 \text{ K})} \right] \\ &= 5.1 \times 10^{-8} \text{ cm/min} \end{aligned}$$

$$\begin{aligned} K_1'' &= 1 \times 10^{-5} \text{ cm/min} \exp \left[\frac{-100000 \text{ J/mol}}{(8.314) (400)} \right] \\ &= 8.7 \times 10^{-9} \text{ cm/min} \end{aligned}$$

$$K'' = \frac{K}{S_e \rho_c} \quad \frac{A_2}{A_1} = \frac{\eta_2 [K_2 / S_e \rho_c] S_A \rho_c (1-E) v/v}{\eta_1 [K_1 / S_e \rho_c] S_A \rho_c (1-E) v/v}$$

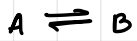
$$\begin{aligned} A_2 &= 2.4 A_1 = 2.4 \times (2.08 \times 10^{-3} \text{ cm}^{-3}) \\ &= 0.005 \end{aligned}$$

$$A_2 = \frac{-\ln \left(\frac{C_A}{C_{A0}} \right) (1-E) \rho_c}{w}$$

$$w = \frac{-\ln (C_A / C_{A0}) (1-E) \rho_c}{A_2}$$

$$\begin{aligned} w &= \frac{-\ln (0.5) (1-0.4) (2.5)}{0.005} \\ &= 207.9 \text{ g} \end{aligned}$$

②



$$\mu = 0.02 \text{ cm}^2/\text{s}$$

$$D = 0.15 \text{ cm} = 0.15 \times 10^{-2} \text{ m}$$

$$u = 9 \text{ cm/s}$$

$$S_{A,C} = 60 \text{ cm}^2/\text{g}$$

$$D_e = 10^{-2} \text{ cm}^2/\text{s}$$

$$K' @ 300\text{K} = 0.01 \text{ cm}^3/\text{s} \cdot \text{g}$$

$$-r_A' = K' C_A$$

$$Sh = 100 Re^{1/2}$$

$$Re = \frac{u d_p}{\mu} = \frac{(9 \text{ cm/s}) (0.15 \text{ cm})}{(0.02 \text{ cm}^2/\text{s})} = 67.5$$

$$K_c = \frac{Sh D_{AB}}{d_p} = \frac{(821.6) (10^{-2} \text{ cm}^2/\text{s})}{0.15 \text{ cm}} = 54.7 \text{ cm/s}$$

$$\text{PBR: } \int \frac{dF_A}{dW} = \int K_{eff} C_A S_{A,C}$$

$$\ln \left[\frac{C_A}{C_{A0}} \right] = -\frac{W}{W_0} K_{eff} S_{A,C}$$

$$W = \frac{-\ln \left[\frac{C_A}{C_{A0}} \right] W_0}{K_{eff} S_{A,C}}$$

$$= \frac{\ln [(1 - 0.6)] (10000 \text{ cm}^3/\text{s})}{(1.6 \times 10^{-4} \text{ cm/s}) (60 \text{ cm}^2/\text{g})}$$

$$= 914462 \text{ g}$$